

MySQL Window Functions - 10 Practice Queries

Functions covered: ROW_NUMBER(), RANK(), DENSE_RANK(), FIRST_VALUE(), LAST_VALUE()

Use the following employees table for all questions.

emp_id	emp_name	department	salary	hire_date
1	Arun	IT	90000	2024-01-10
2	Bala	IT	80000	2024-02-15
3	Charan	IT	80000	2024-03-20
4	Divya	HR	75000	2024-01-12
5	Esha	HR	70000	2024-04-01
6	Farhan	HR	70000	2024-05-05
7	Gokul	Sales	95000	2024-02-02
8	Hari	Sales	85000	2024-06-18

Practice Questions

1. Employee sequence by salary - ROW_NUMBER()

Write a query to display emp_name, department, salary, and a unique row number for all employees ordered by salary from highest to lowest. Use ROW_NUMBER().

2. Salary rank across the company - RANK()

Write a query to rank all employees by salary in descending order. Employees with the same salary must receive the same rank, and the next rank should be skipped.

3. Salary rank without gaps - DENSE_RANK()

Write a query to rank all employees by salary in descending order, but do not leave gaps in the ranking when two or more employees have the same salary.

4. Row number inside each department - ROW_NUMBER()

Write a query to assign a row number to employees separately inside each department. Within every department, the highest-paid employee must receive row number 1.

5. Department-wise salary rank - RANK()

Write a query to rank employees by salary within each department. If two employees in the same department have the same salary, they must share the same rank.

6. Top 2 distinct salaries in each department - DENSE_RANK()

Write a query to return only employees who belong to the top 2 distinct salary levels in their department. Use DENSE_RANK() in a subquery or CTE.

7. Highest-paid employee name per department - FIRST_VALUE()

Write a query that displays every employee along with the name of the highest-paid employee in that employee's department. Use FIRST_VALUE().

8. Lowest-paid employee name per department - LAST_VALUE()

Write a query that displays every employee along with the name of the lowest-paid employee in that department. Use LAST_VALUE() with an explicit window frame so the result is correct for all rows.

9. Compare first and last salary in each department

Write a query that displays emp_name, department, salary, highest salary in the department, and lowest salary in the department using FIRST_VALUE() and LAST_VALUE(). Order the window by salary DESC and use the correct frame.

10. Find the 2nd highest distinct salary employees - DENSE_RANK()

Write a query to return all employees whose salary is the second-highest distinct salary in the entire company. Do not use LIMIT. Use DENSE_RANK().

Quick Function Reminder

Function	Behavior
ROW_NUMBER()	Unique sequence: 1, 2, 3, 4 even for ties
RANK()	Ties share rank; gaps appear: 1, 2, 2, 4
DENSE_RANK()	Ties share rank; no gaps: 1, 2, 2, 3
FIRST_VALUE()	Returns the first value from the ordered window/frame
LAST_VALUE()	Returns the last value from the window/frame; usually requires an explicit full frame

Important: These are practice questions only; solutions are intentionally not included so students can attempt them independently.